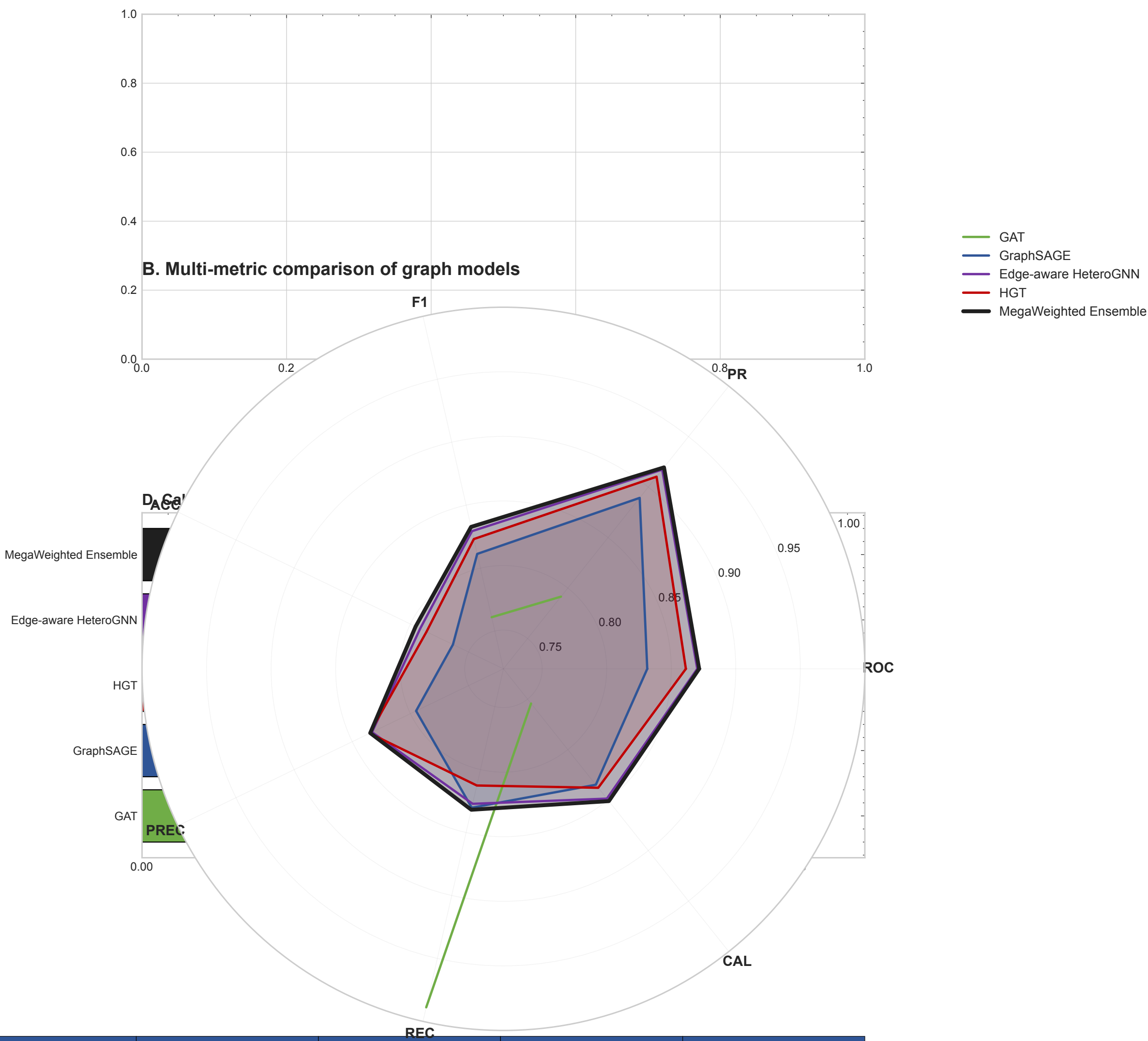
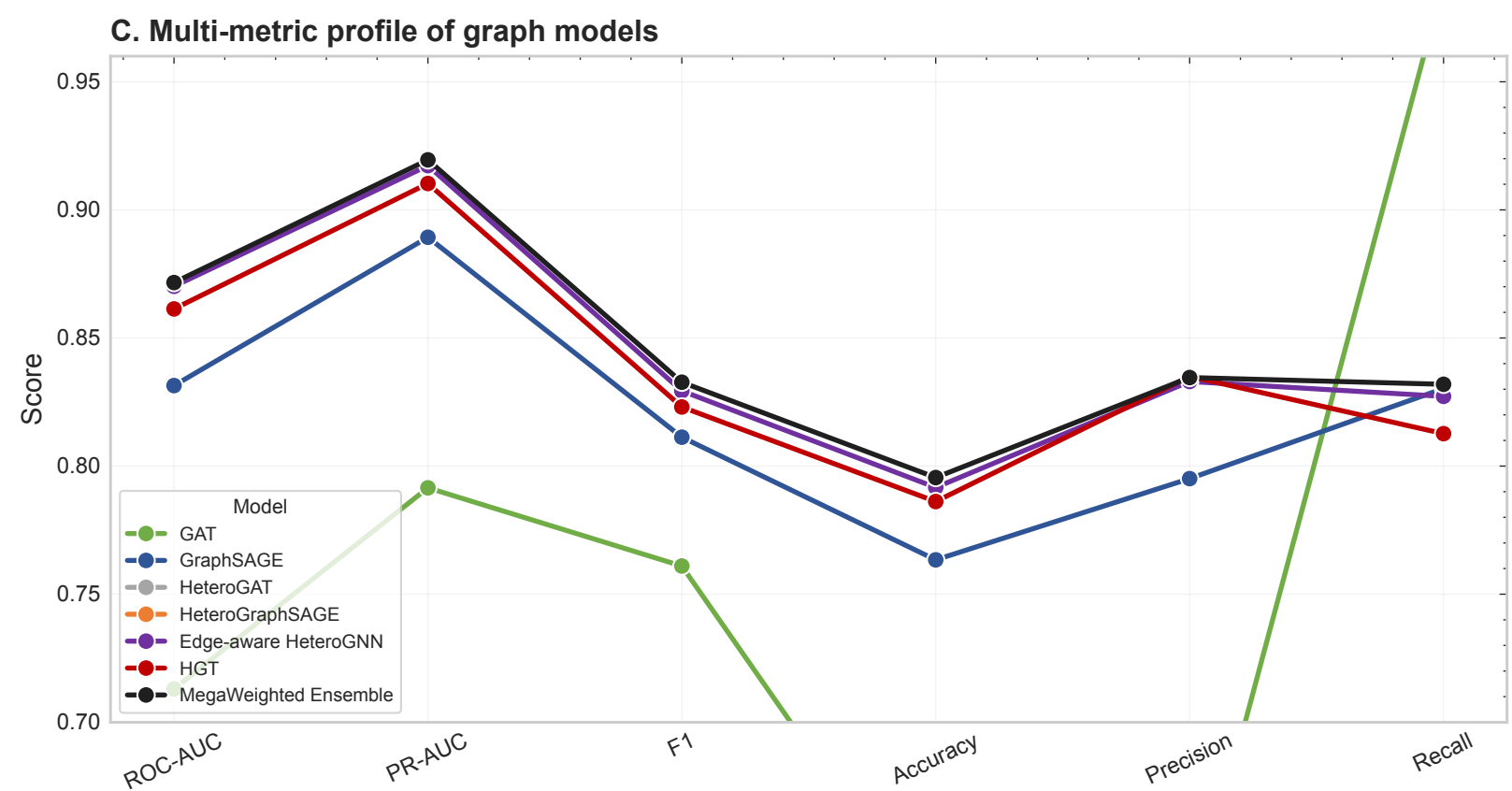
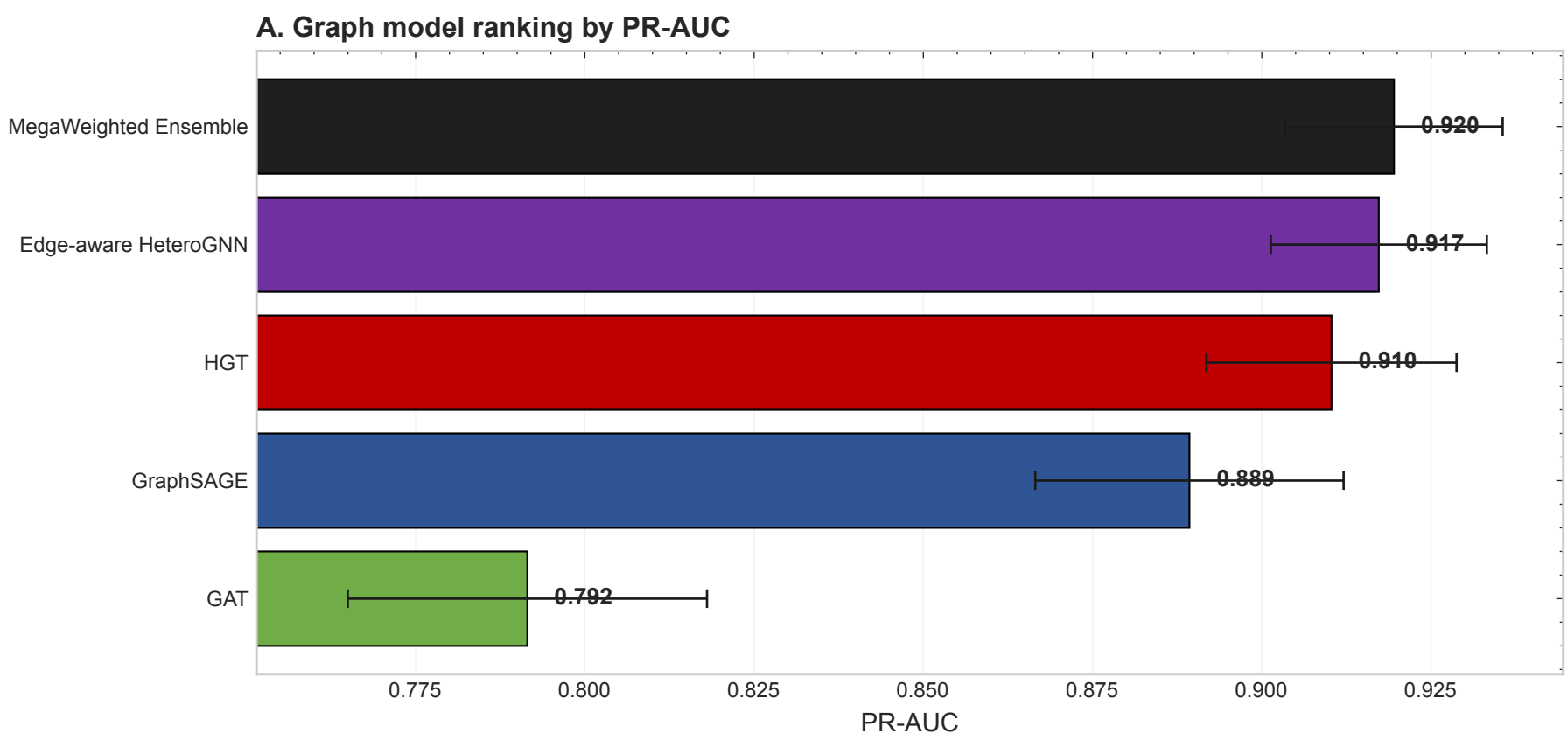


Figure 6. Graph neural network and ensemble benchmark



**E. Graph model performance summary**

| Model                 | ROC-AUC | PR-AUC | F1    | Accuracy | Precision | Recall | Brier |
|-----------------------|---------|--------|-------|----------|-----------|--------|-------|
| GAT                   | 0.713   | 0.792  | 0.761 | 0.619    | 0.619     | 0.989  | 0.246 |
| GraphSAGE             | 0.831   | 0.889  | 0.811 | 0.763    | 0.795     | 0.830  | 0.165 |
| Edge-aware HeteroGNN  | 0.870   | 0.917  | 0.829 | 0.792    | 0.833     | 0.827  | 0.151 |
| HGT                   | 0.861   | 0.910  | 0.823 | 0.786    | 0.835     | 0.813  | 0.162 |
| MegaWeighted Ensemble | 0.872   | 0.920  | 0.833 | 0.796    | 0.835     | 0.832  | 0.149 |